# Data Dictionary

Group-Based Trajectory Modeling, Two Continuous Outcomes (Censored Normal)

**File:** continuous_2outcomes.sas  Traj2 (released core)  **Updated:** 14 October 2026

## 1. Overview

This file implements group-based trajectory modeling (GBTM) for one or two continuous outcomes with bounded support, using a censored-normal (Tobit) likelihood fit by maximum likelihood in PROC NLMIXED. It supports single-outcome fits and a joint two-outcome (multi-trajectory) model that shares one latent-class structure. The joint model uses a warm-start pipeline: each outcome is fit separately first, and those estimates seed the joint fit.

Sourcing the file defines the macros and runs nothing. Set `TRAJ2_RUN_DEMO = 1` before the `%include` to also run the built-in demo pipeline.

## 2. Datasets

| NAME | DESCRIPTION |
|---|---|
| BASE_FILE_SRS | Wide input table: one row per subject, repeated measures per outcome across T time points, plus time-index and cap columns. |
| sim_long / sim_wide | Long and wide outputs of the optional simulator ( `%sim_data_cont` ). |
| work.<output> | ParameterEstimates from `%nlmixed_1` or `%nlmixed_MultiTraj` . The name is whatever you pass to the `output=` argument. |
| nlm_fix_T1&class<br>nlm_fix_T2&class | The `output=` names the built-in demo uses for the two single-outcome fits. Not fixed names. |
| nlm_2y_starting | Stacked single-outcome estimates (alpha rows dropped) that seed the joint fit. |
| nlm_fix_T1_T2&class | The `output=` name the demo uses for the joint two-outcome fit. |
| parameter | Estimates reshaped to one row for prediction and plotting. |
| data_pred | BASE_FILE_SRS crossed with parameters for posterior computation. |
| pred_membership_y | Posterior class-membership probabilities per subject. |
| avg_y1 / avg_y2 | Posterior-weighted observed means per class (via PROC IML). |
| pred / data_plot | Predicted and merged predicted-vs-observed data for plotting. |
| avg_membership | Averaged posterior class-membership probabilities. |

## 3. Global Macro Variables (Step 0)

| NAME | EXAMPLE | DESCRIPTION |
|---|---|---|
| TRAJ2_RUN_DEMO | 0 | 1 runs the built-in demo pipeline when the file is sourced; 0 defines the macros only. Default is 0. With 0 the log prints "Demo pipeline skipped" and no model is fit. |
| USE_SIM | 1 | 1 = run the simulator; 0 = use your own BASE_FILE_SRS. |
| T | 12 | Number of time points. |
| class | 5 | Number of latent classes. |
| order_model | 3 | Polynomial order (1=linear, 2=quadratic, 3=cubic). |
| equal | T | T = equal residual sigma across classes; F = class-specific. |
| max_values | 10 10 … 10 | Censoring caps, one per time point (length T). |

| NAME | EXAMPLE | DESCRIPTION |
|---|---|---|
| y1vars | QHH1-QHH12 | Outcome 1 repeated-measure variables. |
| y2vars | QINP1-QINP12 | Outcome 2 repeated-measure variables. |
| idvar | BENE_ID | Subject identifier variable. |
| tvars | quar1-quar12 | Time-index variables (values 1..T). |

## 4. Key Variables

| NAME | TYPE | DESCRIPTION |
|---|---|---|
| BENE_ID | Num | Subject identifier; carried to output for linkage. |
| QHH1-QHH12 (y1vars) | Num | Outcome 1 repeated measures, one column per time point. |
| QINP1-QINP12 (y2vars) | Num | Outcome 2 repeated measures. |
| quar1-quar12 (tvars) | Num | Time indices, values 1..T. |
| CAP1-CAP&T | Num | Per-time censoring caps loaded from max_values. |
| X[T] | Array | Time predictor inside the model. |
| Y1[T], Y2[T] | Array | Outcome arrays inside the model. |
| alpha0_[class] | Num | Class-membership intercept (class A fixed to 0). |
| beta[outcome]_[class][k] | Num | Degree-k polynomial coefficient, given outcome and class. |
| sigma[outcome]_[class] | Num | Residual SD; class suffix omitted when equal_sigma=T. |
| mu[class][outcome][T] | Array | Class- and outcome-specific latent mean over time. |
| PI[class][outcome][T] | Array | Per-time LOG density contribution for that class and outcome. Set to 0 at a missing time point, which contributes 1 to the product and skips that time point. |
| PROD[class][outcome] | Num | Running sum of the log contributions across time for that class and outcome. |
| e[outcome]_[class] | Num | Residual (observed minus predicted). |
| pie_[class] | Num | Marginal class-membership probability. |
| pinumer_[class] / pideno | Num | Numerator and denominator of the class-probability softmax. |
| post_[class] | Num | Posterior probability of class membership per subject. |
| ll_latclass | Num | Log-likelihood contribution passed to the MODEL statement. |

## 5. Macros

| NAME | PURPOSE |
|---|---|
| %sim_data_cont | Optional simulator: two censored-normal outcomes. |
| %add_caps | Loads CAP1-CAP&T from the max_values list. |
| %build_base_file_srs | Assembles BASE_FILE_SRS (simulated or user-supplied). |
| %starting_value_alpha | Neutral starting values for mixing intercepts (A fixed to 0). |
| %starting_value_beta_sigma | Starting values for polynomial coefficients and sigma. Optional `int_lo=`, `int_hi=` and `sigma0=`. See the note below. |
| %bounds_alpha / %bounds_sigma | Parameter bounds for stable estimation. |
| %initiation_universal | Declares shared time and outcome arrays. |
| %initiation | Declares per-class, per-outcome arrays for fitting. |

| NAME | PURPOSE |
|---|---|
| `%initiation_pred` | Declares per-class, per-outcome arrays for prediction. |
| `%model` | Builds the polynomial mean for each class. |
| `%residual` | Computes residuals per class. |
| `%float_control` | Caps extreme residuals for numerical stability. |
| `%Prob_cnorm` | Censored-normal log-density contribution per time point. |
| `%Class_Membership` | Softmax class-membership probabilities. |
| `%LogLike / %LogLike_multi` | Single-outcome and joint mixture log-likelihood. |
| `%Posterior` | Posterior class probabilities per subject. |
| `%Pred_cnorm` | Censored-mean prediction per class. |
| `%plot_prep` | Builds predicted-vs-observed plots and the posterior summary. |
| `%nlmixed_1` | Fits a single-outcome model. Uses `tech=dbldog`. |
| `%nlmixed_MultiTraj` | Fits the joint two-outcome model. Uses `tech=dbldog`. |

Three further macros, `%_traj2_demo_default`, `%_traj2_demo_build` and `%_traj2_demo_pipeline`, are internal. They read `TRAJ2_RUN_DEMO` and run the demo. Do not call them directly.

**Starting values.** Leaving `int_lo=` and `int_hi=` blank on `%starting_value_beta_sigma` gives every latent class an identical intercept of 0. That is a symmetric start for a finite mixture and the classes have nothing to separate them. Supplying the two bounds spreads the class intercepts evenly across the plausible outcome range, which is what the published validation runs do. For an outcome censored to [0,100] with residual SD near 6: `int_lo=10, int_hi=45, sigma0=6`.

## 6. Notes and Assumptions

- Missing time points are allowed (.) and skipped in the likelihood, valid under missing-at-random.
- Censoring is recognized by exact equality of an outcome to its cap; the cap vector must match the true measurement limits, so validate against the sample max.
- Class labels A, B, C, ... come from `%let class_names`; changing the count must be reflected in the `class=` and `LC=` arguments.
- `equal_sigma=T` estimates one sigma across classes; F estimates one per class.
- The joint fit depends on starting values: use the warm-start pipeline (single fits, stack, joint) and set `int_lo=` and `int_hi=` to reduce non-convergence.
- `%float_control` clips residuals to plus or minus 8 residual SDs before the density is evaluated, to stop exp(PROD) underflowing to zero. This is a deviation from the exact likelihood in MethodsX Section 2.1.1.